

Understanding phi-psi preferences of amino acids

Hitesh Kandarpa¹, Atharva Mandar Phatak²,

¹Department of Biotechnology, Indian Institute of Technology Madras, Chennai, Tamil Nadu, India

²Department of Biotechnology, Indian Institute of Technology Madras, Chennai, Tamil Nadu, India

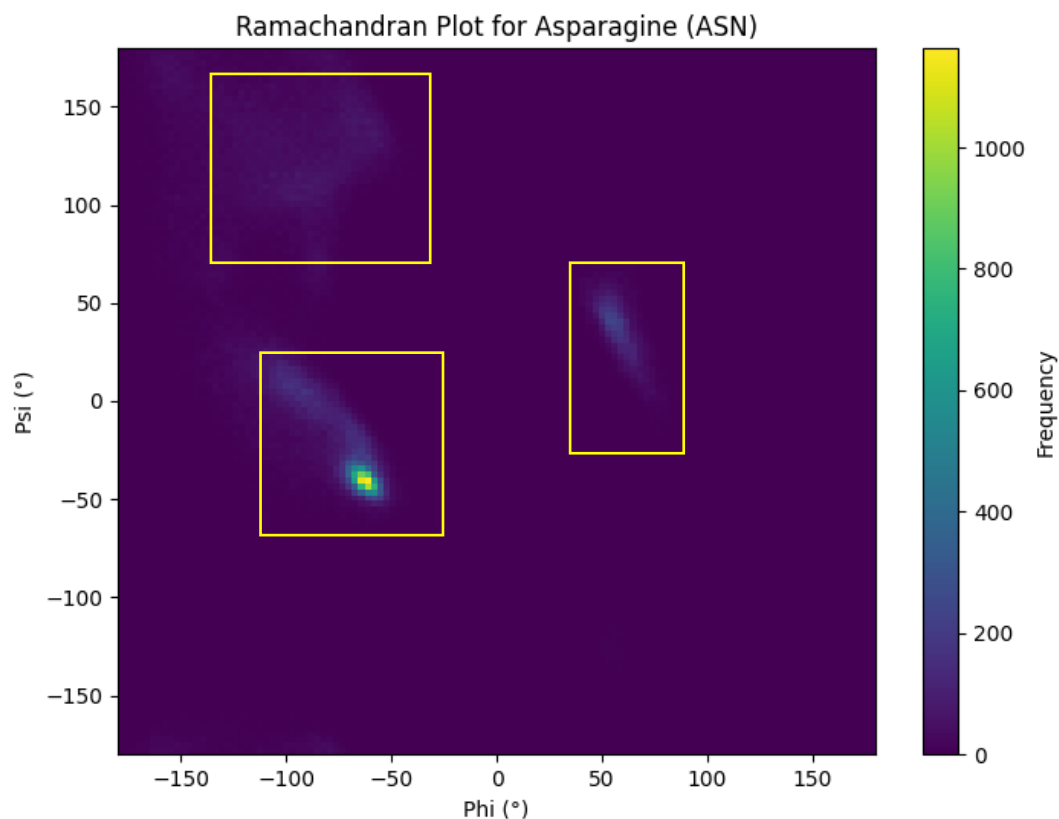
Course: BT5340 | Protein Folding and Stability

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Introduction

Asparagine is among less than a handful of amino acids capable of reacting with saccharides (referring to the biologically active L-asparagine stereoisomer). It has been found that asparagine in potatoes reacts with glucose through a Maillard reaction to form acrylamide, a neurotoxin and potential carcinogen. Its side chain can react to form N-glycosidic bonds with oligosaccharides, establishing a prevalent role in the glycosylation of proteins during the latter phase of protein synthesis. Asparagine is involved specific protein functions, such as antibodies, collagen assembly, and also metabolic control of cell functions in nerve and brain tissue. [Ref]

How does the distribution of phi-psi of residue X appear in the Ramachandran plot? In other words, bin the occurrences in $3^\circ \times 3^\circ$ cells and plot the distribution

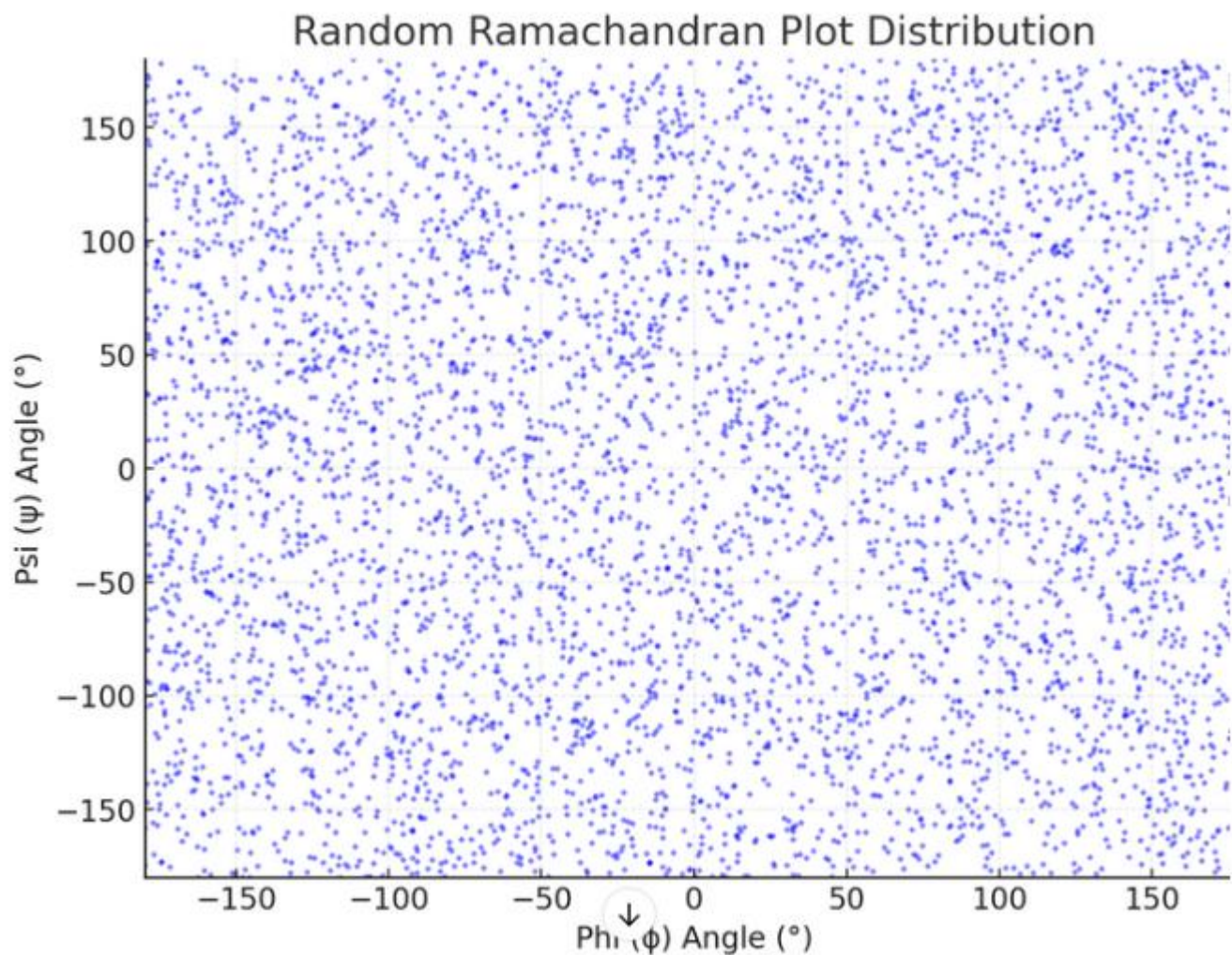


The primary cluster (high density, high concentration (yellow regions)) is located near $(\phi, \psi) = (-60^\circ, -50^\circ)$, which corresponds to the **right-handed α -helix region**. This shows asparagine's ability to participate in α -helical structures likely due to favourable backbone hydrogen bonding and side-chain flexibility. A secondary, less prominent cluster is

observed near $(\phi, \psi) = (60^\circ, 50^\circ)$, which corresponds to the **left-handed α -helix region**. A sparse cluster is visible around $(\phi, \psi) = (-80^\circ, 120^\circ)$, corresponding to the **β -sheet region**. Absence/ Low Density of points is seen in the **disallowed** regions.

How different is the distribution from a random distribution?

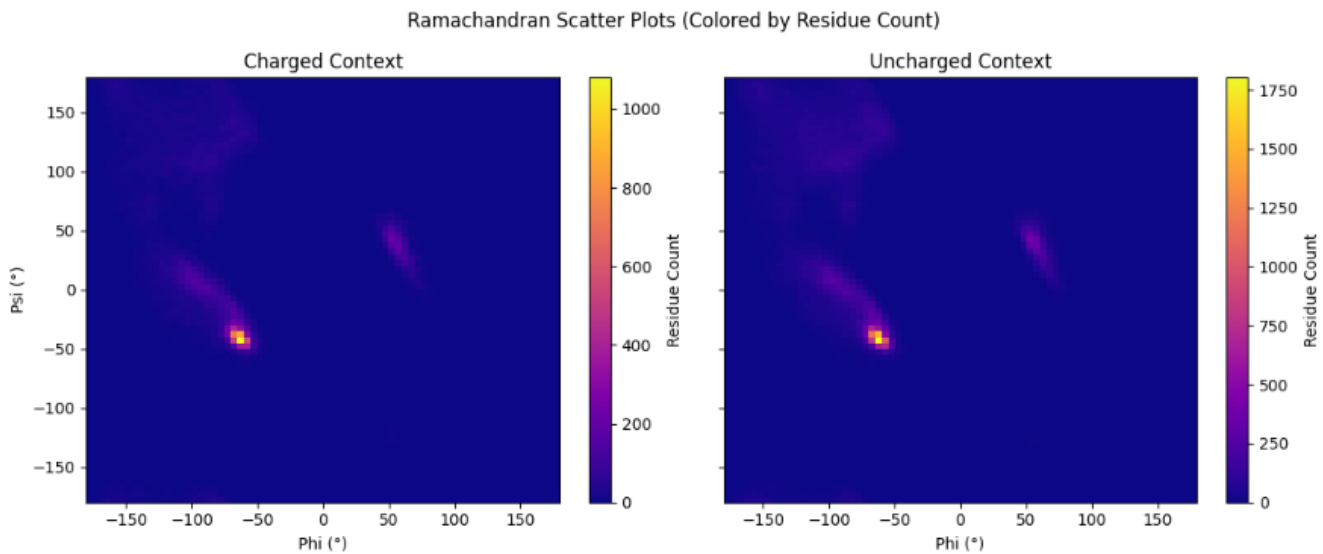
A **random** (ϕ, ψ) distribution would look roughly **uniform** across the entire Ramachandran space (-180° to $+180^\circ$) on both axes). However, the observed plot shows dense clustering in α -helix regions indicating strong structural preferences as peptide backbone cannot adopt all (ϕ, ψ) combinations without incurring large steric clashes.



Explain the rationale for deviations, if any, you observe from a conventional Ramachandran plot seen in Biochemistry textbooks.

Asparagine's side chain contains **an amide group capable of forming hydrogen bonds** with nearby residues or water molecules. Asparagine side chains form rigid hydrogen-bonded ladders in HET-s(218–289) fibrils, stabilizing β -sheet structures. [Ref]. It is also observed that Asparagine's side chain frequently forms **hydrogen bonds** with the backbone NH of residue $n+2n+2$, stabilizing tight turns (e.g., type I β -turns). This "**pseudo-turn**" motif occurs 2.5 $\times$ more often in asparagine than other residues. [Ref] Its side chain is moderately bulky but not excessively large (e.g., compared to tryptophan), allowing it to access allowed regions while avoiding sterically disallowed areas. Unlike glycine (high flexibility) or proline (rigid backbone), asparagine has intermediate flexibility, allowing it to adopt both α -helical and β -sheet conformations but avoiding disallowed regions. Asparagine residues are often found near the beginning of alpha-helices as α -turns and α -turn motifs, and in similar turn motifs, or as amide rings, in beta sheets. This positioning allows asparagine to "**cap**" **hydrogen bond networks**, satisfying interactions that would otherwise be fulfilled by the polypeptide backbone [Ref].

Consider a 7-residue segment of sequence 'a-b-c-X-d-e-f' with X being at the centre of this segment. Extract all such sequences from the database of protein structures you have created. Do you see any differences in the phi-psi preference of X if there is at least one charged residue either side of the central residue X? If so, rationalize the answer



The **overall distribution of ϕ - ψ angles** appears **highly similar** in both cases, with no significant shifts in the preferred conformational regions. Although densities of few regions are slightly varying, there is **no clear evidence** that charged residues strongly influence the backbone conformations of Asparagine. Both plots show clusters in roughly the same ϕ - ψ regions, hence Asparagine's backbone flexibility remains relatively unchanged regardless of nearby charged residues.

Rationalisation:

a. Asparagine's Intrinsic Rigidity:

Asn has a **polar, uncharged side chain (-CONH₂)** that can form hydrogen bonds, but it does not carry a formal charge. Unlike charged residues (Asp, Glu, Lys, Arg), Asn does not experience strong electrostatic interactions that could significantly restrict or alter its backbone conformation.

b. Minimal Electrostatic Influence on the Backbone:

Charged residues in the local sequence could theoretically affect the backbone by **forming salt bridges or repelling nearby residues**. However, we conclude that these effects do **not strongly propagate** to Asn's ϕ - ψ angles.

Do you see differences in the phi-psi distribution of residue X (compared to the distribution generated from the first question) if the 7-residue segment is present in a helix or a strand?

We were not able to get this plots.

Extras:

Warning: Undefined array key "option" in /var/www/html/pisces/PISCES_ChoseInputPage.php on line 3

PISCES: A Protein Sequence Culling Server

Step 1: Choose your desired thresholds

Maximum pairwise percent sequence identity:	25
Minimum resolution (X-ray and EM):	0.0
Maximum resolution (X-ray and EM):	2.0
Maximum R-value (X-ray only):	0.25
Minimum chain length:	40
Maximum chain length:	1000
Include X-ray entries?	☒ Yes ☐ No
Include cryo-EM entries?	☐ Yes ☒ No
Include NMR entries?	☒ Yes ☐ No
Include chains with chain breaks? (clicking "No" will reduce number of chains by ~20%)	☒ Yes ☐ No
Include chains with missing residues due to disorder? (clicking "No" will reduce number of chains by 50-75%)	☒ Yes ☐ No
Write file of pairwise sequence identities of returned list?	☐ Yes ☒ No

Step 2: Submit your job

results from PISCES cullpdb_pc25.0_res0.0-2.0_len40-1000_R0.25_Xray+Nmr_d2025_03_01_chains9858

External

Inbox x



roland.dunbrack@fcc.edu

to me ▼

Your thresholds for culling the PDB

Resolution : 0.0 - 2.0

R-factor : 0.25

Sequence length : 40 - 1000

Sequence percentage identity: <= 25.0

X-ray entries: Included

EM entries: Excluded

NMR entries: Included

Chains with chain breaks: Included

Chains with disorder: Included

Number of chains: 9858

Download result files: http://dunbrack.fccc.edu/pisces/users/be21b009@smail.iitm.ac.in/cullpdb_pc25.0_res0.0-2.0_len40-1000_R0.25_Xray+Nmr_d2025_03_01_chains9858.tar